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Department of Computer Science and Engineering (CSE-2)

A.Y:2026-27 **IV B.Tech Odd Semester** **Y23 Batch**

Engineering Capstone Project – I

Abstract Proforma **Date:** __/08/26

Batch Number: 58 **Domain:** Visual Intelligence Systems **Project Use case no:** DSCI-41	**Project Title:** Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video	**Guide Name:** Dr. T. Praveen **Guide emp. Id:** 5495

Project Abstract: Intelligent Accurate vehicle counting from roadside CCTV cameras is a fundamental requirement for Intelligent Transportation Systems (ITS), enabling traffic monitoring, congestion analysis, and urban transportation planning. However, reliable vehicle perception remains challenging due to frequent occlusions, varying vehicle scales, and complex urban traffic scenarios. In this work, a real-time vehicle perception framework is developed by integrating the YOLOv8 object detector with the ByteTrack multi-object tracking algorithm to detect, classify, and track multiple vehicle categories, including cars, buses, trucks, and motorcycles, from continuous CCTV video streams. The framework is trained and evaluated using the Barbados Traffic Analysis dataset. YOLOv8 performs vehicle detection and classification, while ByteTrack maintains consistent object identities across consecutive frames, thereby reducing tracking failures caused by temporary occlusions and missed detections. The proposed framework generates continuous class-wise traffic volume as time-series data, which serves as the primary input for subsequent traffic congestion forecasting models. Establishing this perception baseline enables systematic evaluation of traffic forecasting performance and facilitates future analysis of the impact of adverse weather conditions on intelligent transportation systems.

To be filled by the Guide (during abstract submission):

Student IDs:	**Student Attendance**	**Student's Understanding Level on Project** (V. Good: 10-8M Good: 7-6M Average: 5-4M Poor: 3-0M)
2300031316		
2300033502		
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2300030938		

Guide remarks:

Guide signature